

Mixed cropping of narrow-leaved lupin with oat enhances grain-yield stability across five European locations

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1 Abstract

Background & aims: Pulse x cereal mixtures are promoted to stabilise yield and reduce fertiliser dependence, yet robust multi-environment evidence for narrow-leaved lupin (NLL, *Lupinus angustifolius*) × oat (*Avena sativa*) mixed cropping remains scarce. We tested whether NLL × oat mixtures maintain productivity while enhancing grain yield stability.

Methods: A randomised complete block design (four blocks) evaluated three NLL and three oat varieties grown as pure stands and as all nine pairwise mixtures in a replacement design (NLL 90%, oat 10% of pure-stand density). The experiments ran for 13 environment-years across five European locations. Productivity was summarised with land equivalent ratio (LER) and crop performance ratio (CPR). Yield stability was assessed using (i) the coefficient of variation (CV), (ii) Finlay-Wilkinson regression slope (b_1) and Eberhart–Russell deviation variance (S_{di}^2), and (iii) additive main effects and multiplicative interaction-derived weighted average of absolute scores (AMMI-WAAS). Mixed-effects models were used to assess mixed cropping effects.

Results: Across environments, LER did not differ from unity ($LER = 1$) whether fertilised or unfertilised oat monocultures were the reference. By contrast, total CPR and oat specific CPR consistently exceeded 1 (mean CPR = 5.3; CPR_N = 4.6), indicating system level gains driven by the oat component without penalising NLL. While CVs did not separate treatments, many mixtures and pure stands were environmentally responsive, but the most stable mixtures combined near average responsiveness with very low variation and sensitivity to the environmental conditions (Bison × Probor and Troll × Jowisz). Lastly, fertilised oat monocultures were more stable than unfertilised counterparts.

Conclusions: Stability gains arose from reduced environmental sensitivity of the mixture rather than the individual variety, underscoring the need to evaluate mixtures directly rather than inferring stability from monocultures. These results support deploying pulse-centric cereal × legume mixtures as a pragmatic option to buffer climate variability while sustaining protein-rich production.

Keywords: mixed cropping, *Lupinus angustifolius*, *Avena sativa*, LER, CPR, Finlay–Wilkinson, AMMI, WAAS, yield stability, genotype × environment.

2 Introduction

Global demand for protein-rich plants is growing while climate volatility continues to undermine the year-to-year reliability of monocultures (Ismail et al., 2020; Raseduzzaman and Jensen, 2017). Mixed cropping of cereals with grain legumes (pulses) has re-emerged as a promising strategy to improve yield stability, decrease fertiliser dependence and improve resource-use efficiency (Kumawat et al., 2022; Raseduzzaman and Jensen, 2017; Rodriguez et al., 2020; Tang et al., 2021). Narrow-leaved lupin (NLL, *Lupinus angustifolius* L.) is among the few pulses that combine high seed protein with the ability to thrive on nutrient poor soils (Lopez-Bellido and Fuente, 1986); however, its sole-crop yield is notoriously unstable (Reckling et al., 2018). Oat (*Avena sativa* L.) offers complementary morphology, rapid early growth and serves as a cereal break crop (e.g. Danso et al., 1993; Pálmason et al., 1992). Integrating the two species in mixed cropping can therefore address both agronomic resilience and nutritional targets at farm and food-system level, respectively.

Numerous studies on genotype $\times$ cropping-system interactions have demonstrated that performance gains from cereal $\times$ pulse mixtures depend on both the mixed varieties and the environment in which they were evaluated (e.g. (Demie et al., 2022)). Meta-analyses consistently report average land-equivalent ratios (LER) between 1.2 and 1.3 for cereal-pulse mixtures, indicating a median 24% land-use advantage over pure stands (Demie et al., 2022). Grain-yield stability has been assessed with both static and dynamic metrics; in many cases, mixtures outperformed monocultures, yet the magnitude and drivers of this stability remain unclear (Raseduzzaman and Jensen, 2017; Weih et al., 2021).

Fertiliser regimes of cereals in pure stands are pivotal for interpreting mixed-cropping metrics because both LER and crop performance ratio (CPR) are normalised by monoculture benchmarks (Hauggaard-Nielsen and Jensen, 2001; Liu et al., 2023). In formal terms, the CPR compares observed mixed cropping output to the output expected from monoculture yields at the same sowing proportions (Zustovi et al., 2024). As a consequence of fertilising cereals in pure stands and the resulting increase in yield, the denominator of the cereal LER and CPR raises, which results in lower LER and CPR-metrics for the same mixed crop. Though rarely investigated, reference treatment fertilisation exhibits significant effects on the mixed cropping performance metrics (Baxevanos et al., 2020). Additionally, medium to high fertiliser levels were shown to stabilise cereal yield (Silva et al., 2016). Thus, the choice of baseline (fertilised vs. unfertilised) pure cereal stands directly affects mixed-cropping advantages in terms of productivity and stability metrics (Li et al., 2020; Mead and Willey, 1980).

Despite this theory, robust multi-environment evidence for NLL $\times$ oat mixtures is scarce. Previous experiments have often been confined to single locations, or ad-hoc varietal choices, leaving unresolved (i) whether crop-level indices such as the CPR validate classical LER assessments, (ii) how canopy height difference between the varieties influence the agronomic performance, (iii) whether the expected grain-yield stability gains of mixed cropping materialise across contrasting European agroclimates and (iv) how fertilisation of the pure reference cereal crop affects yield indices and stability metrics in mixed stands. Addressing these gaps is critical, since pulses are pivotal to climate-change mitigation strategies and nitrogen-fertiliser reduction programmes (Rahman et al., 2022).

Accordingly, we expect

- Mixed stands to achieve LER values above 1, indicating superior land use compared with pure stands
- Total CPR and the oat CPR to exceed unity (CPR above 1), reflecting an overall mixed cropping advantage
 - The CPR and LER to be relatively smaller if those metrics are compared with fertilised rather than unfertilised pure stands of oats
- NLL $\times$ oat mixed cropping to increase both static and dynamic grain-yield stability across the five European locations relative to their corresponding pure stands and
 - The gain in grain-yield stability of mixed stands to be relatively smaller if that gain is compared with fertilised rather than unfertilised pure stands of oats.

3 Methods

3.1 Varieties

Three branching spring NLL and three spring oat varieties were chosen based on the breeders' fact sheets and the variety list of the German variety institute (Bundessortenamt, 2021). Branching, spring-NLLs and early maturing spring oat varieties were selected to increase the chances of coincidental ripening. The selected, NLL varieties were

- Lunabor, bred by Saatzucht Steinach in Germany
- Probor, bred by Saatzucht Steinach in Germany
- Jowisz (synonym: Jupiter), bred by HR Smolic in Poland

The selected early maturing spring oat varieties were:

- Bison, Hauptsaaen für die Rheinprovinz GmbH, Germany, medium tall
- Lion, Nordsaat Saatzucht GmbH, Germany, medium tall
- Troll, Saatzucht Bauer GmbH & Co. KG, Germany, short.

3.2 Treatments and experimental design

The NLL and oat varieties were cultivated as spring crops in pure and mixed stands. The pure oat stands were fertilised according to the local guidelines (Supplemental materials - Table S1). In the mixed stands, each NLL variety was co-cultivated with each oat variety in a replacement design: the NLLs were sown at 90% of their pure stand's density, whereas the oats, due to their high competitiveness and the focus on the NLL crop, were sown at 10% of their pure stand's density (Böhm et al., 2008; Kato-Noguchi et al., 1994) (Table 1). The sowing amounts were adapted to the germination rate.

Treatment	Stand	Fertilisation*	Sowing density** (plantsm ⁻²)	Amount (kg ha ⁻¹)
Bison	pure	no	300 - 400	168 - 224
Lion	pure	no	300 - 400	121 - 161
Troll	pure	no	300 - 400	110 - 147
Bison	pure	yes	300 - 400	168 - 224
Lion	pure	yes	300 - 400	121 - 161

Table 3.1

Treatments of the study: narrow-leaved lupin and oat varieties, the stand (pure versus mixed), fertilisation, sowing ratios, respective sowing densities and corresponding amount of seeds per hectare. For the two numbers presented in the mixed treatments, the first one refers to the narrow-leaved lupins, the second one to the oats. The sowing density and fertilisation of the oat crop varied depending on the local recommendations.

Treatment	Stand	Fertilisation*	Sowing density** (plantsm ⁻²)	Amount (kg ha ⁻¹)
Troll	pure	yes	300 - 400	110 - 147
Lunabor	pure	no	120	220
Probor	pure	no	120	170
Jowisz	pure	no	120	214
Lunabor × Bison	mixed	no	108 / 30 - 40	198 / 17 - 22
Lunabor × Lion	mixed	no	108 / 30 - 40	198 / 12 - 16
Lunabor × Troll	mixed	no	108 / 30 - 40	198 / 11 - 15
Probor × Bison	mixed	no	108 / 30 - 40	153 / 17 - 22
Probor × Lion	mixed	no	108 / 30 - 40	153 / 12 - 16
Probor × Troll	mixed	no	108 / 30 - 40	153 / 11 - 15
Jowisz × Bison	mixed	no	108 / 30 - 40	193 / 17 - 22
Jowisz × Lion	mixed	no	108 / 30 - 40	193 / 12 - 16
Jowisz × Troll	mixed	no	108 / 30 - 40	193 / 11 - 15

*Fertilisation was performed according to local guidelines (Supplemental materials - Table S1).

**Novi Sad: 400 plants · m⁻², Bad Vilbel and Gross Lüsewitz: 350 plants · m⁻², Reckenholz and Seegräben: 300 plants · m⁻²

The fields of this two-year study were arranged as a randomised complete block design (RCBD), consisting of four blocks. To lower the border effect to a minimum, each treatment was sown in three adjacent plots and only the middle plot was scored and harvested. The size of each harvested plot was 9m².

3.3 Locations, husbandry and weather data

The experiments were conducted in Germany (Bad Vilbel and Gross Lüsewitz), Serbia (Novi Sad) and in Switzerland (Seegräben and Reckenholz) and were chosen based on their differing latitude, longitude, soil type (Supplemental materials - Table S1) and weather conditions (Table 2). In Switzerland, the experiments were performed in 2022 and 2023, while in Serbia and Germany, they were conducted in 2022, 2023 and 2024. All fields were ploughed and harrowed before sowing. The NLL seeds were inoculated with peat inoculum containing the rhizobium bacterium *Bradyrhizobium lupinus*. The inoculum was produced by Legume Technology Ltd, Nottinghamshire, UK. The same batch of inoculum was used in all years and stored dry and under exclusion of humidity at 10°C. After the inoculation, every 9m² plot was sown individually in rows with 18 cm spacing. The plots were sown with a plot sowing tractor and RTK-GPS-orientation. Spring-tine and shallow inter-row cultivation was performed for the mechanical weeding. Chemical weed control was applied only at Gross Lüsewitz in 2022 and 2024 using the pre-emergence herbicide Stomp Aqua; at all other sites, weeds were controlled mechanically or manually, or no weeding was required (Supplementary Materials - Table S1). The fertilised pure oats were fertilised during late tillering, before the elongation stage, according to local guidelines (Supplemental materials - Table S1). At Bad Vilbel, following organic production principles, horn grit was used as fertiliser.

Location and year	Mean temperature (°C)	Precipitation sum (mm)
Bad Vilbel 2022	13.4	174
Bad Vilbel 2023	13.1	293
Bad Vilbel 2024	13.8	402

Location and year	Mean temperature (°C)	Precipitation sum (mm)
Gross Lüsewitz 2022	11.8	234
Gross Lüsewitz 2023	11.8	258
Gross Lüsewitz 2024	13	367
Novi Sad 2022	22.7	75
Novi Sad 2023	17.8	234
Novi Sad 2024	17.4	175
Seegräben 2022	14.1	504
Seegräben 2023	13.8	500
Reckenholz 2022	17.7	244
Reckenholz 2023	17.4	362

3.4 Harvest and post-harvest analyses

Final plant canopy heights of the NLLs and oats were scored before harvesting. The crops were harvested with a plot combine harvester. The samples were then dried to a moisture content below 13%. In mixed treatments, the oats were separated from the NLLs with grain cleaners, where a 6mm oval-cut sieve was deployed. The cleaned samples were weighted to measure the partial and total yield.

3.5 Statistical analysis

3.5.1 Data preparation and calculations

Based on the primary data, productivity metrics were calculated to assess the the LER of the mixture based on unfertilised oats (LER) and fertilised oats (LER_N) in pure stands, respectively. Further, the CPR of each crop based on unfertilised oats in pure stands (CPR_{lupin}, CPR_{oat}), and based on fertilised oats in pure stands (CPR_{oat_N}) was determined, respectively, and the total CPR based on unfertilised oats in pure stands (CPR) and fertilised oats in pure stands (CPR_N) was calculated, respectively (Zustovi et al., 2024):

$$LER = \frac{Yield_{lupin, mixed}}{Yield_{lupin, pure}} + \frac{Yield_{oat, mixed}}{Yield_{oat, unfertilised, pure}}$$

$$LER_N = \frac{Yield_{lupin, mixed}}{Yield_{lupin, pure}} + \frac{Yield_{oat, mixed}}{Yield_{oat, fertilised, pure}}$$

$$CPR_{crop} = \frac{Yield_{crop, mix}}{p_{crop} \times Yield_{crop, unfertilised, pure}}$$

$$CPR_{oat_N} = \frac{Yield_{oat, mix}}{p_{oat} \cdot Yield_{oat, fertilised, pure}}$$

$$CPR = CPR_{oat} + CPR_{lupin}$$

Table 3.3

Weather data of the location and year in the cultivation seasons 2022, 2023 and 2024 (13 environment-years), retrieved from the closest official weather stations (the distance of the weather station to the location was between 0.25 and 5 km). Data were retrieved daily and the average temperature as well as the sum of precipitation across the location- and year-specific cultivation period were calculated.

$$\text{CPR_N} = \text{CPR}_{\text{oat_N}} + \text{CPR}_{\text{lupin}}$$

Where:

$\text{Yield}_{\text{crop, mixed}}$ is the yield of the crop in mixture ($\text{t} \cdot \text{ha}^{-1}$),

$\text{Yield}_{\text{crop, pure}}$ is the yield of the same crop in pure stand ($\text{t} \cdot \text{ha}^{-1}$), where unfertilised and fertilised specifies if the pure reference stands of oats were fertilised or not

p_{crop} is the proportion of the crop's sowing density in the mixture relative to its pure stand, and

crop represents either NLL or oat.

Those metrics allowed to compare the mixed cropping system's land use and crop performance relative to their monocropping equivalents. By definition, the LER in pure stands is 1, the CPR of one crop is 1 if a crop does not benefit nor suffer from mixed cropping, and the total CPR of two crops is 2 if the two crops do not interact with each other.

3.5.2 Modelling

The response variables were analysed using linear mixed-effects models. Four model-classes M1-M4 were designed to suit the target-variables (Supplemental materials - Table S2). The treatment, year, location, and their interaction effects were modelled as fixed effects and the block within a field as a random effect. The models were fitted using the `lme4` package within the R programming language (Bates et al., 2015; Team, 2025). Inferences were calculated using the `lmerTest` package, which provides the Satterthwaite degree of freedom adjustment for the type-III ANOVA and the `contest`-function, which was used to test linear contrasts (Kuznetsova et al., 2017). The contrast sum parametrisation was used to allow the interpretability of the regression coefficients to the population mean, and partially fitted intercepts were used for each group in case of the M2 and M4 models (R syntax: 'y~0+group'). Post-hoc pairwise comparisons ($\alpha = 0.05$) were performed to assess varietal differences.

3.5.3 Robustification

To mitigate the influence of outliers, the outlier weights were automatically adapted using the weights from the analogous model derived by the `rlmer` function of the `robustlmm` package (Koller, 2016). Target outliers were selected by applying the bounded influence function of Huber loss. This application of the robustification is indicated in the Supplemental materials, Appendix A (`robustify = TRUE`).

3.5.4 Static grain-yield stability

To assess treatment-level differences in the static grain-yield stability, the coefficient of variation (CV) of the partial NLL, partial oat and the total yield for each treatment were calculated. The CV was computed using the standard deviation and mean of total yield, following the implementation in the metan package (Olivoto, 2020). A permutational multivariate analysis of variance (PERMANOVA) was applied for differences among treatments, years and locations on Bray–Curtis distances to compare treatments pairwise (9999 permutations; strata = block) (Arbizu, 2020; Oksanen et al., 2001). P-values were adjusted by Benjamini–Hochberg across all pairs.

3.5.5 Dynamic grain-yield stability

The dynamic grain-yield stability across environments (location $\times$ year) was assessed using the univariate Finlay–Wilkinson regression approach, also known as joint regression (Finlay and Wilkinson, 1963). The analysis was implemented performing the regression-based stability analysis according to Eberhart and Russell (1966) (Supplemental materials - Formulas 1 and 2). The calculated slope coefficient, b_1 , represents treatment responsiveness to the environment. A b_1 slope coefficient close to 1 indicates average responsiveness: the treatment performs proportionally to environmental productivity. Values of b_1 above 1 reflect a more responsive or opportunistic treatment, showing above-average yield under favourable conditions but reduced performance in less productive environments, whereas b_1 below 1 denote a stable or conservative treatment that maintains relatively consistent yield across environments. The calculated Eberhart–Russell deviation variance S_{di}^2 at low levels indicates that the treatment’s response to the environment is highly predictable and stable (consistent across environments), whereas a high variance indicates erratic or inconsistent performance across environments, i.e. high dispersion of the data points around the slope coefficient b_1 .

To analyse the treatment-by-location interactions, and to quantify the grain-yield stability based on a multivariate model, the additive main effects and multiplicative interaction (AMMI) model was applied (Gauch, 1992; Zobel et al., 1988). The AMMI model decomposed the observed total yield response Y_{ij} of treatment i in location j into additive main effects for treatment and location, plus multiplicative interaction effects captured by principal components axes (IPCA):

$$Y_{ij} = \mu + G_i + E_j + \sum_{k=1}^N \lambda_k \gamma_{ik} \delta_{jk} + \varepsilon_{ij} \quad (4)$$

Where:

μ is the grand mean,

G_i and E_j are the treatment and location main effects,

λ_k is the singular value of the k^{th} interaction principal component axis (IPCA),

γ_{ik} and δ_{jk} are the treatment and location scores for the k^{th} IPCA, and

ε_{ij} is the residual term.

Based on the AMMI-model, the weighted average of absolute scores (WAAS) was computed to summarise multi-dimensional interaction effects into a single stability index, following Olivoto et al. (2019) as implemented in the metan R package (Olivoto (2020)):

$$WAAS_i = \sum_{k=1}^N w_k |IPCA_{ik}| \quad (5)$$

Where w_k is the proportion of the interaction sum of squares explained by the k^{th} IPCA.

The WAAS summarises the multi-dimensional treatment-by-location interaction from the AMMI model into a single index of stability. Lower WAAS values indicate smaller absolute interaction effects and thus higher grain-yield stability, while higher WAAS values denote stronger environmental sensitivity. Together, the b_1 , S_{di}^2 and WAAS indices capture the facets of yield stability: b_1 describes the directional response to the location, S_{di}^2 the dispersion around the direction of the response, and WAAS quantifies the magnitude of interaction-driven variability across locations and years, i.e. the treatment's sensitivity to environmental changes.

4 Results

4.1 Grain yield

In Reckenholz, the most productive location, median grain yield of the pure fertilised oat stands ranged between 4.9 and 5.4 t · ha⁻¹. Pure NLL stands produced 2.8 – 3.6 t · ha⁻¹, while the NLL x oat mixtures reached 3.7–4.8 t · ha⁻¹. Gross Lüsewitz and Seegräben exhibited similar yield to Reckenholz in pure fertilised oat stands, however, the NLL yield and total yield were lower in Gross Lüsewitz ($p < 0.001$). Reckenholz and Seegräben exhibited the highest NLL yield in pure stands compared with the other locations ($p < 0.001$)(Figure 1).

The total yield of Bad Vilbel and Novi Sad were lower across all observed treatments in the year 2022 than the total yield of Reckenholz, Seegräben or Gross Lüsewitz ($p < 0.001$), whereas in 2023, only the total yield of Gross Lüsewitz was significantly lower than the yield recorded in Novi Sad, Reckenholz and Seegräben (Supplemental materials - Table S3). Further, the yield of all NLL and oat varieties in pure stands ($p < 0.001$) and the partial yield of Jowisz × Troll in mixed stands differed significantly ($p < 0.05$). Across all locations the average yield of pure NLL and oat stands were not significantly different from the average yield across all mixtures. Regarding the proportion of the partial NLL yield in mixtures, Bad Vilbel and Seegräben exhibited higher NLL than oat proportions of 62% and 59% across all years, respectively, whereas the proportions of the NLL yield in mixtures amounted to 35%, 29% and 24% in Gross Lüsewitz, Novi Sad and Reckenholz, respectively ($p < 0.001$). A strong year effect was apparent, resulting in an elevated proportion of the NLL yield by 16% in 2022 and 7% in 2023 across all mixtures, respectively. Additionally, an oat-interaction effect was detected, resulting in an increased NLL proportion by 5% if the NLL were mixed with Troll, and a decreased proportion by -4.6% if the NLL were mixed with Lion. All locations exhibited strong year-by-year variation of the NLL proportion (Supplemental materials - Appendix A).

4.2 Grain-yield performance

The LER_N assessed across all locations with regards to fertilised oats as pure stands, years and treatments did not differ from unity (LER_N = 1) on a treatment, nor location level (Supplemental materials - Appendix A). Additionally, no difference could be detected between any pairs of mixed stand treatments or locations. Though the LER with regards to unfertilised oats was higher than the LER_N, it still did not differ from unity (LER = 1). The total CPR with regards to fertilised and unfertilised oats as reference pure stands was elevated above unity ($E = 2$) for all mixed stands, in absence of any treatment effect, amounting to an average CPR of 5.3 ($p < 0.001$), and CPR_N = 4.6 ($p < 0.001$), respectively, driven by the high oat CPR (Supplemental materials - Appendix A). The CPR_N was significantly lower than the recorded CPR in all mixed treatments but Troll × Lunabor and Troll × Jowisz ($p < 0.05$). None of the CPR or CPR_N values exhibited significant differences between treatments across all locations and years. Similar to the total CPR, the CPR_{oat} was above unity ($E = 1$), with regards to both the fertilised (CPR_{oat_N}) and the unfertilised (CPR_{oat}) oats as pure reference stands ($p < 0.001$). The CPR_{oat} was significantly larger than the CPR_{oat_N} for all mixed

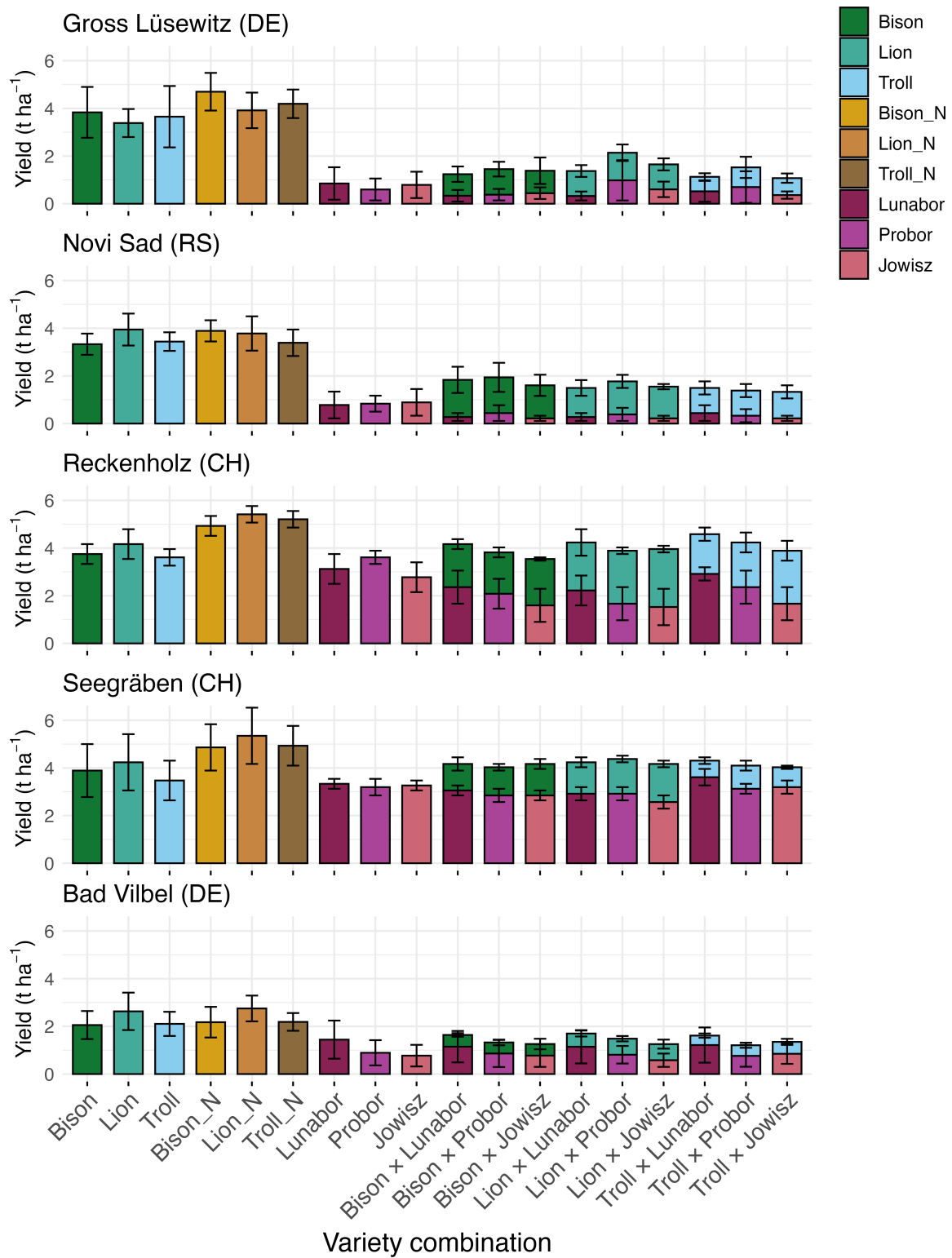


Figure 4.1: Grain yield overview of pure and mixed stands by location and the location's country code in brackets. Fertilised oats are shown in yellow-brown hues. For all graphs, the median value is represented by the bar height, and the variability (median absolute deviation) by the error bars. Data were pooled across all years. Bars with a single colour represent pure stands. In mixed stands, the lower bar represents the partial lupin values, the upper bar the partial oat values.

Table 4.1

Overview of all grain-yield stability metrics summarised across all locations on a treatment basis. CV = Coefficient of variation, b_1 = Finlay–Wilkinson slope, S_{di}^2 = Eberhart–Russell deviation variance, WAAS = weighted average of absolute scores. Letters correspond to the significance grouping of the pair-wise post-hoc tests ($p < 0.05$) adapted for multiple testing. Variety_N represents fertilised pure oat stands.

Bison and Lion treatments ($p < 0.05$), whereas the mixed Troll treatments were agnostic to the fertilisation regime of their pure reference stands. Post-hoc comparisons unveiled significantly higher CPR_oat and CPR_oat_N in three cases of Lion, compared with Troll, where Lion had shown a CPR exceeding the Trolls CPR by 0.8 - 0.9 ($p < 0.01$). However, these effects were restricted to specific mixed cropping comparisons (e.g. Lion $\times$ Probor compared with Troll $\times$ Probor) but did not hold true globally for both oat varieties across all paired mixtures. Conversely to the CPR of oats and the total CPR, the CPR of NLLs did not differ from unity ($E = 1$), and there was no effect of variety, location or year effects.

4.3 Grain-yield stability

4.3.1 Static grain-yield stability

The static grain-yield stability, expressed as the coefficient of variation (CV) calculated on a treatment basis across all environment-years unveiled no differences in the coefficient of variation across partial NLL, oat yield or the total yield (Table 3). However, trends were visible for the yield variation compared with the treatments' mean yield, the variations was larger in pure stands of NLLs and some mixed stands, relative to the majority of the mixed stands and pure (un-)fertilised oat stands. Further, fertilised and unfertilised oats did not differ significantly with regards to their CV.

Treatment	CV total yield (not significant)	b_1 (responsiveness)	S_{di}^2 (dispersion)	WAAS (sensitivity)
Bison	17	0.68a	0.24a	0.66a
Lion	16	0.63a	0.14a	0.58a
Troll	17	0.57a	0.3 a	0.80a
Bison_N	18	1.1 a	0.21a	0.35a
Lion_N	16	1.14b	0.05b	0.08b
Troll_N	15	1.2 a	0.09a	0.15b
Lunabor	31	0.95a	0.22a	0.44a
Probor	35	1.05a	0.23a	0.48a
Jowisz	27	1.01a	0.19a	0.38a
Bison $\times$ Lunabor	17	1.08b	0.004b	0.11b
Bison $\times$ Probor	18	0.98c	0.03b	0.04b
Bison $\times$ Jowisz	18	0.91c	0.06 b	0.08b
Lion $\times$ Lunabor	15	1.05b	0.03 b	0.21b
Lion $\times$ Probor	16	1.14a	0.09 a	0.34a
Lion $\times$ Jowisz	15	1.05b	0.01 b	0.1b
Troll $\times$ Lunabor	17	1.21b	0.03 b	0.2b
Troll $\times$ Probor	27	1.23b	0.06 b	0.35a

Treatment	CV total yield (not significant)	b_1 (responsiveness)	S_{di}^2 (dispersion)	WAAS (sensitivity)
Troll $\times$ Jowisz	24	1.03b	0.0005b	0.067b

4.3.2 Dynamic grain-yield stability

The Finlay-Wilkinson regression coefficients (b_1) measuring the responsiveness of the total yield to environmental impact was significantly elevated ($b_1 > 1$) in pure stands of fertilised Lion, and the mixed stands of Bison $\times$ Lunabor, Lion $\times$ Lunabor, Lion $\times$ Jowisz and Troll $\times$ Probor, whereas Bison $\times$ Probor and Bison $\times$ Jowisz exhibited a buffering capacity ($b_1 < 1$) to the environmental influence ($p < 0.005$). All pure stands of NLLs and oats, other than fertilised Lion, and the mixed stands of Lion $\times$ Probor and Troll $\times$ Jowisz exhibited average environmental responsiveness. The Eberhart–Russell deviation variance S_{di}^2 was highest in pure stands of NLLs, and lowest in mixed treatments (Table 3). The pure stand of fertilised Lion and Lion $\times$ Probor both combined high responsiveness b_1 with low S_{di}^2 dispersion, indicating that both treatments reliably and consistently respond to their environmental conditions.

The AMMI model unveiled two groups according to the WAAS-scores: the sensitivity to environmental conditions of pure unfertilised oat stands was higher than that of mixed stands (Figure 2, Table 3), further the AMMI-Model unveiled significant location and treatment effects but no interaction thereof (Supplemental materials - Table S4). The WAAS-scores indicated the mixture’s higher stabilities across all environment-years, due to their weaker sensitivity to the location effect. The WAAS scores also indicated a tendency towards greater stability in fertilised pure oat stands than in unfertilised pure oat stands, although this difference was not statistically significant. Across the assessed interaction principal components axes (IPCA), IPCA1 distinguished between the two groups of locations Novi Sad and Gross Lüsewitz (negative loadings) versus Bad Vilbel, Reckenholz and Seegräben (positive loadings) (Figure 3). Additionally, IPCA1 clearly distinguished the pure stands of NLLs (positive loadings) from the oats (negative loadings). By contrast, IPCA2 distinguished the fertilised pure stands of oats (negative loadings) from the unfertilised pure stands of oats. With regards to the factor loading of IPC1 and IPC2, the two mixtures Lion $\times$ Jowisz and Troll $\times$ Jowisz plotted closest to the origin (0,0). These two treatments interacted least with the locations along IPCA1 and IPCA2, and were thus exhibiting a more stable performance across all five locations than treatments positioned at higher Euclidean distance to the origin (Supplemental materials - Table S4).

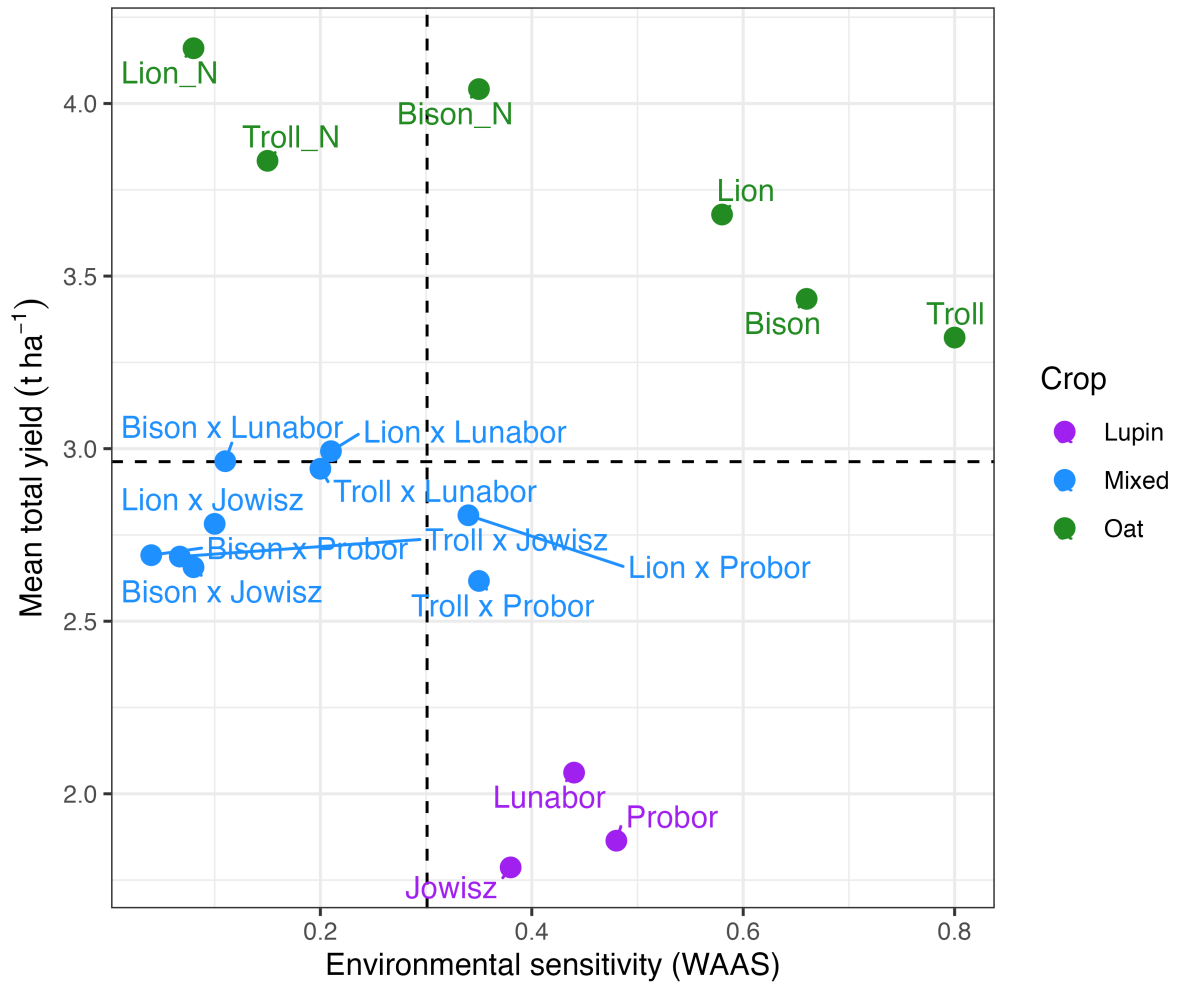


Figure 4.2: Mean total yield plotted against WAAS, an index of environmental sensitivity, for pure and mixed stands across 2022–2024 (2022–2023 for the Swiss sites). Treatments in the upper left quadrant combine high yield with low sensitivity, whereas points further to the right indicate lower grain-yield stability. Dashed lines mark the median values of both axes.

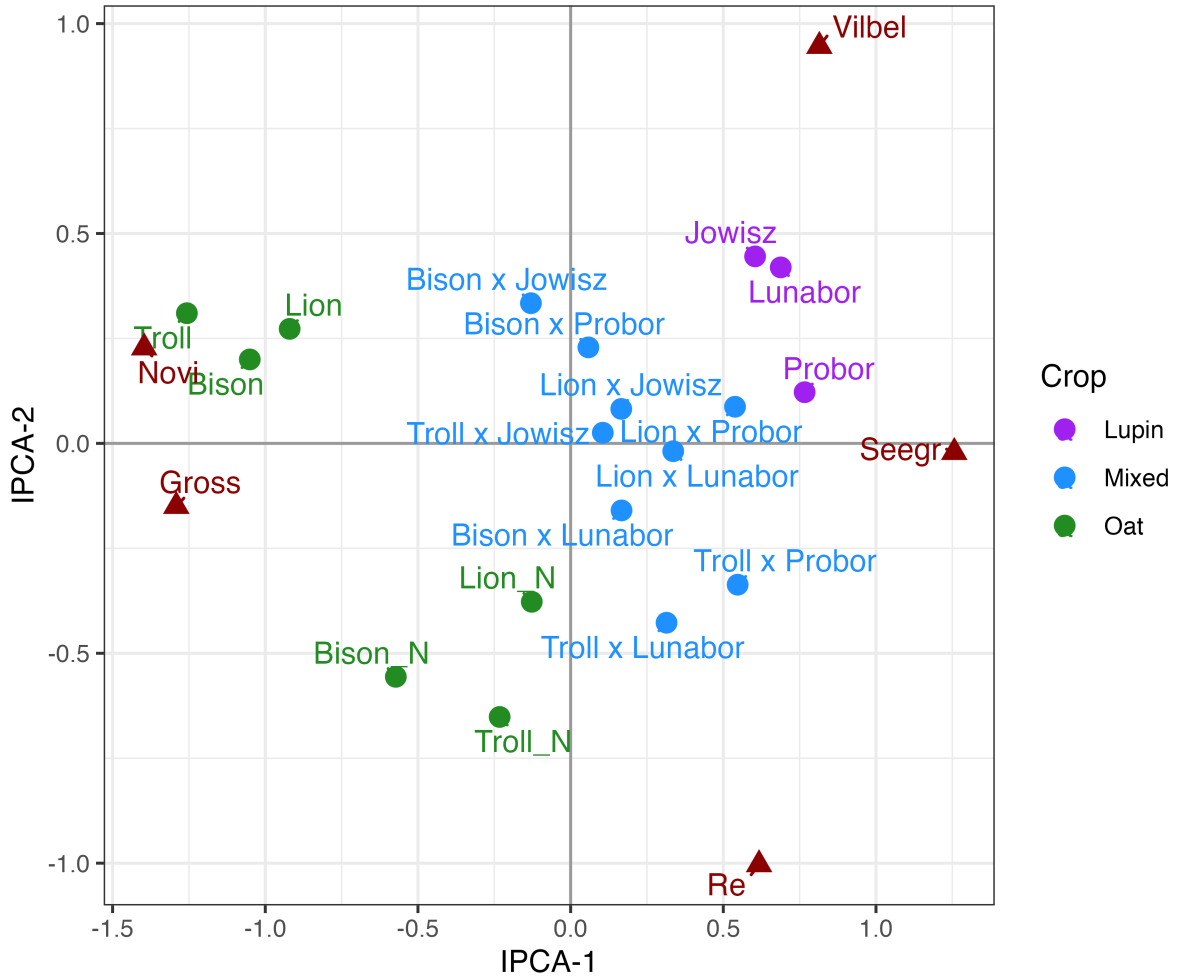


Figure 4.3: The additive main effects and multiplicative interaction (AMMI) biplot of the treatment (circles) and location (triangles) scores on the first two interaction principal component axes for total grain yield in pure and mixed stands. The closer a treatment is positioned to the origin ($x,y = 0, 0$), the higher was its grain-yield stability. The following location acronyms were used: Gross - Gross Lüsewitz (2022 - 2024), Novi-Novu Sad (2022 - 2024), Vilbel - Bad Vilbel (2022 - 2024), Re - Reckenholz (2022 - 2023) and Seegr - Seegräben (2022 - 2023).

5 Discussion

The multi-environment evaluation of NLL $\times$ oat mixed cropping demonstrated that total NLL grain yield was maintained across five European locations, while several metrics showed an improved temporal and spatial grain-yield stability relative to the pure stands. The yields recorded in the present study were broadly within the range reported for other European pulse $\times$ cereal mixed-cropping systems, although direct comparisons should be interpreted with caution because the species combinations, environments, and management conditions differed between studies. For oat-based intercrops, mean yields of around $3.1 \text{ t} \cdot \text{ha}^{-1}$ have been reported for oat-vetch systems in Germany and Sweden and for oat-pea systems in Germany, which is comparable to the yield range observed at Reckenholz and Seegräben in the present study (Engbersen et al., 2021; Munz et al., 2025). By contrast, the remaining locations showed lower total yields across both pure and mixed stands, consistent with the yield levels reported by Wang et al. (2012) for lentil-oat intercrops under temperate European conditions. Because these lower yields occurred consistently across all pure and mixed treatments within the respective locations, they are more plausibly attributed to environmental than to varietal effects, in line with Brooker et al. (2024) showing that intercrop performance remained strongly modulated by rainfall and crop management. Overall, the yield levels observed in the present study are in good agreement with the broader European synthesis by Bedoussac et al. (2015), who reported a mean intercrop grain yield of $3.3 \text{ t} \cdot \text{ha}^{-1}$ across 58 pulse $\times$ cereal mixtures.

The productivity indices allow for a clear distinction between land use efficiency and crop performance: The LER did not differ significantly from unity, but this should not be interpreted as an absence of mixed-cropping benefit. Rather, it indicates that, on a land-area basis, the mixtures on average matched the reference pure stands, while the component responses within the mixtures were asymmetric: This asymmetry was captured by the CPR: oat CPR and total CPR were consistently above unity, whereas the CPR of the NLL remained unchanged, showing that the oat crop captured additional resources without depressing the NLL performance. This interpretation is consistent with Zustovi et al. (2024), who showed that mixed cropping indices evaluate different facets of performance and respond differently to crop ratio and total sowing density. Accordingly, in a pulse-centric replacement design such as the present one, the CPR, an indicator accounting for sowing proportion, is more sensitive than the LER to gains of the minority component.

In the present study, yield maintenance was achieved through a flexible contribution of the two partners rather than through a fixed proportion: the NLL proportion in mixture ranged from 24% in Reckenholz to 62% in Bad Vilbel. Similar shifts in pulse $\times$ cereal contribution as a function of weather and location conditions have been reported for oat-vetch mixtures, underlining that the agronomic value of intercropping lies in partner compensation across environments rather than in a constant botanical composition (Bedoussac et al., 2015; Brooker et al., 2024; Pużyńska et al., 2021). This beneficial effect of mixed cropping in stabilising the grain yield can be explained by niche complementarity and mutual yield buffering; the NLL and oat captured resources not used by the companion crop, thus increasing resource use efficiency, as supported by recent theoretical work that niche complementarity underlies mixture advantages (Demie et al., 2022; Hauggaard-Nielsen et al., 2001; Jensen, 1996). Additionally, in the pulse-centric NLL $\times$ oat mixed cropping system, a strong buffering capacity has been reported, where the respective

yields of NLL and oat are correlated negatively, thus jointly buffering yield fluctuations compared to pure stands (Schlup et al., 2026). This pronounced niche complementarity effect in oat, as reflected by its elevated CPR, coupled to the yield buffering of both species thus explains the improved grain-yield stability of the mixtures across all environment-years and their lower response to environmental stress or stimuli.

Using either fertilised or unfertilised oat pure stands as the reference for LER led to the same conclusion: LER did not differ between baselines. This insensitivity to the fertiliser regime is reflected by substantial spatiotemporal variance in yield across all environment-years and the low sowing proportion of oat in the mixtures, which diminishes the weight of the oat component in the summed LER, thus limiting statistical power. Different mixed cropping indices respond differently to crop ratio and total sowing density (Zustovi et al., 2024), explicitly LER did not elucidate a fertiliser effect, whereas the CPR did: Both the total CPR and oat-specific CPR were higher when mixtures were benchmarked against unfertilised oat stands, rather than fertilised oat stands, revealing a fertiliser-mediated gain in the performance of pure oat stands. Because CPR explicitly scales yields by the sowing proportion of each species, it provides a proportion-corrected measure and is therefore more sensitive to management effects under imbalanced mixtures than the LER. These findings are consistent with the prior report of Baxevanos et al. (2020) that using unfertilised pure stands as references can yield larger estimates of mixed cropping advantage than using fertilised references.

While the static metric (CV) did not detect significant differences in grain-yield stability, the two dynamic approaches, the Finlay-Wilkinson slope (b_1) and the WAAS, revealed clear contrasts among treatments. In pure stands, fertilised oat Lion combined an above-average responsiveness to environments (slope $b_1 > 1$) with low dispersion around the regression, indicating a predictable response in both favourable and unfavourable conditions. Fertilisation enhanced the stability of oats in pure stands: in the present study, fertilised Lion and Troll showed significantly higher grain-yield stability (smaller WAAS-scores) than their unfertilised counterparts, consistent with the findings of Silva et al. (2016). Most mixed stands exhibited elevated responsiveness to the environmental conditions ($b_1 > 1$), while the most stable mixed treatments exhibited average responsiveness ($b_1 \approx 1$), coupled to low sensitivity to the environmental conditions WAAS. Such stability in pulse $\times$ cereal mixtures is commonly attributed to differential species responses and complementarity that buffer environmental variation and reduce $G \times E$ interaction, resulting in more stable yield amongst selected combination (Raseduzzaman and Jensen, 2017). Across all varieties, the LER remained consistently at its reference level of one, while the oat and total CPR values were consistently above one, highlighting a global mixed-cropping effect of said metrics, which is agnostic to variety-pairing. This reinforces the evidence that the overall mixed-cropping effect outweighs cultivar-specific effects, as reported for barley-pea mixtures and for maize in maize-cowpea mixtures (Hauggaard-Nielsen and Jensen, 2001; Watiki et al., 1993). The pattern of varying grain-yield stability gains across the investigated mixtures demonstrates that elevated grain-yield stability was not only a mixed-cropping effect, but was also depended on the specific pairing of NLL and oat varieties. Comparable variety-specific differences have been reported in other pulse $\times$ cereal mixed crops, where mixture performance varied with the complementarity of the respective partners and their response to resource availability (Weih et al., 2021). Mechanistic work in oat-NLL mixtures further suggests that such variability arises from differences in temporal matching of crop traits, including early vigour, competitive ability, and the timing of resource capture (Engbersen et al., 2021). Acknowledging the general mixed-cropping advantage in grain-yield stability and in parallel focussing on variety-specific interactions and benefits is thus the best way forward to optimise grain-yield stability in pulse-cereal production.

The increased dynamic grain-yield stability observed for mixed cropping in this study extends the findings of Weih et al. (2021) by showing that under a pulse-centric replacement design (NLL $\times$ oat), mixture stability did not only match but surpassed the pure-crop benchmarks on

dynamic grain-yield stability. Furthermore, this pattern re-affirms the findings by Raseduzzaman and Jensen (2017), that replacement-designed mixed crops of cereals and pulses elevate the grain-yield stability across environments. Notably, the mixtures Bison $\times$ Probor and Troll $\times$ Jowisz, emerged as the most stable, even though neither of their corresponding pure stands ranked among the most stable, underscoring that stability must be assessed at the mixture level rather than inferred from respective pure stands. By quantifying stability with complementary dynamic metrics (Finlay–Wilkinson slope, b_1 , and WAAS), the results of this study show that the stability advantage of mixtures is not confined to balanced designs; it also holds true under replacement designs that emphasise the NLL component.

6 Conclusion

This study focusses on the grain-yield performance and stability of pulse $\times$ cereal mixed cropping compared with pure stands between different locations and years. The patterns observed in the present study demonstrate that optimal variety mixtures improve grain-yield stability of both NLL and oat while maintaining crop performance and land use, opening opportunities for tailoring mixtures to agri-environmental risk profiles. Moreover, the robust yield-stability advantage of mixtures across environment-years, which differed six-fold in seasonal precipitation, suggests that mixed cropping can effectively buffer against increasing climate variability, aligning with policy aims to increase the production of plant-based proteins. Hence, strategic variety pairing within mixed cropping emerges as an expedient lever to combine high crop yield and maintain land use performance while improving the production's resilience. These findings strengthen the evidence base for incorporating cereals into pulse-centric cropping strategies to enhance resilience under climate uncertainty and mitigate the pulses' yield instability. Future studies should address additional aspects of mixed cropping, such as economic viability as well as processing and quality of harvested grains.

7 CRediT authorship contribution statement

Schlup Yannik: Conceptualisation, investigation, data assembly, statistical analysis, methodology, project administration, resources, software, visualisation, writing, editing. **Andrea Gallehr, Kathrin Neubeck, Matthias Heinrich Herrmann, Radivoje Jevtić, Selma Schurack, Vesna Župunski:** Conceptualisation, investigation, data assembly, project administration, resources, review and editing. **Lukas Graz:** Statistical analysis, software, visualisation, review. **Johan Six:** Review and editing. **Susanne Vogelgsang:** Conceptualisation, funding acquisition, methodology, project administration, resources, review and editing.

8 Funding

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9 Declaration of competing interest

The authors declare that they have no known personal relationships or competing financial interests that could have influenced the work reported in this paper.

10 Declaration of generative AI and AI-assisted technologies in the writing process.

During the preparation of this work the authors used GPT-4 (OpenAI, California, USA) to improve the structure and efficiency of the code to produce figures and GPT-o3pro (OpenAI, California, USA) in support of writing the abstract, introduction and discussion. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the created content.

11 Data and analytical code

The data and code supporting the findings of this study are available at the project's GitHub repository (<https://github.com/incorey/lupin-oat-grain-yield-stability>) and are permanently archived on Zenodo under DOI <https://doi.org/10.5281/zenodo.XXXXXXX>; all materials are released under the MIT License.

12 Acknowledgments

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13 Supplemental materials

Formula 1 - Eberhard-Russell regression for plot-level data

$$Y_{ij} = \mu_i + b_{1i} I_j + \delta_{ij} + \varepsilon_{ij} \quad (1)$$

Where:

Y_{ij} is the observed yield of the i^{th} treatment in the j^{th} environment,

μ_i is the mean yield of the i^{th} treatment across all environments,

b_{1i} is the Finlay-Wilkinson regression coefficient representing treatment i 's responsiveness to the environments,

I_j is the environmental index, calculated as the mean yield of environment j minus the grand mean across environments,

δ_{ij} is the deviation from the linear response in environment j used to compute the deviation variance S_{di}^2 , and

ε_{ij} is the residual term.

Formula 2 - Eberhart-Russell deviation variance

$$S_{di}^2 = \frac{\sum_{j=1}^E \delta_{ij}^2}{E - 2} \quad (2)$$

Where:

$\delta_{ij} = Y_{ij} - (\mu_i + b_{1i} I_j)$ are the deviations from the regression line and

E are the number of environments.

Location and year	Latitude (°N)	Longitude (°E)	Pre-crop	Pre-pre-crop	Sowing	Weeding 1	Weeding 2	Oat fertilisation (kgNha ⁻¹)	Pest intervention	Harvest	Soil-type
Bad Vilbel 2022	50.2	8.6	-	-	March 29	April 28	May 5	43	-	July 14	Sandy loam
Bad Vilbel 2023	50.2	8.8	-	-	April 06	-	-	40	-	August 1	Sandy loam

Table 13.1

Table S1: Field characteristics and husbandry measures per location and year (environment-year).

Location and year	Latitude (°N)	Longitude (°E)	Pre-crop	Pre-pre-crop	Sowing	Weeding 1	Weeding 2	Oat fertilisation (kgNha ⁻¹)	Pest intervention	Harvest	Soil-type
Bad Vilbel 2024	50.2	8.7	-	-	April 24	-	-	35	-	July 29	Sandy loam
Gross Lüsewitz 2022	54.1	13.3	Lupin	Lolium	March 31	April 5 (Stomp Aqua)		40	-	August 17	Sandy loam
Gross Lüsewitz 2023	54.1	13.3	Lupin	Lolium	April 5	-		40	-	August 22	Sandy loam
Gross Lüsewitz 2024	54.1	13.3	Lolium	Winter rye	March 22	March 27 (Stomp Aqua)		40	-	August 8	Sandy loam
Novi Sad 2023	45.3	19.8	Maize	Winter wheat	May 5	manual	manual	60	Van-tex 60CS against Oulema ssp.	July 28	Loamy chernozem
Novi Sad 2024	45.3	19.8	Soy-bean	Maize	March 30	manual	manual	60	Van-tex 60CS against Oulema ssp.	July 19	Loamy chernozem
Novi Sad 2025	45.3	19.8	Soy-bean	Maize	February 28	manual	manual	60	Van-tex 60CS against Oulema ssp.	July 1	Loamy chernozem
Reckenholz 2022	47.4	8.5	Silage maize	Flower strip	March 4	March 21 (mechanical)	April 5 (mechanical)	27.5	-	July 12	Loamy clay
Reckenholz 2023	47.4	8.5	Silage maize	Artificial pasture	February 16	February 22 (mechanical)	April 4 (mechanical)	30	-	July 10	Loamy clay
See-gräben 2022	47.3	8.9	Silage maize	Winter wheat	March 10	March 17 (mechanical)	March 31 (mechanical)	27.5	-	July 19	Sandy loam

Location and year	Latitude (°N)	Longitude (°E)	Pre-crop	Pre-crop	Sowing	Weeding 1	Weeding 2	Oat fertilisation (kgNha ⁻¹)	Pest intervention	Harvest	Soil-type
Seegräben 2023	47.4	8.9	Silage maize	Winter barley	February 22	March 1 (mechanical)	April 5 (mechanical)	30	-	July 19	Sandy loam

Model	Characteristics
M1	This model answered if the response variable differed in mixtures and pure NLLs (or analogously oats) across all varieties. It performed an ANOVA using the NLL/oat-variety and the binary variable stand = mixed. This model considered each species separately.
M2	M2 answered if the response variable differed in mixtures and pure NLLs or oats. Here, this was achieved by computing the linear contrasts of each pure NLL/oat variety with the mean of its respective mixtures. This model considered each species separately.
M3	On the subset of only mixed plots, M3 performed a two-way (mixed) ANOVA with the factors oat variety and NLL variety (adjusting for location, year, and block). This model considered both species together. Further, M3 performed post-hoc comparisons of the treatments and locations.
M4	M4 answered the question, if the estimated value of the treatments with mixed stands (e.g. the land equivalent ratio) was significantly different from an expected value (e.g. LER = 1) after a Bonferroni correction (with the respective 12 coefficients).

Treatment	Gross 2022	Gross 2023	Gross 2024	Novi 2022	Novi 2023	Novi 2024	Re 2022	Re 2023	Seegr 2022	Seegr 2023	Vil-bel 2022	Vil-bel 2023	Vil-bel 2024
Bison	3.83	2.30	4.85	2.95	3.33	4.38	4.51	3.33	5.35	3.06	2.81	0.77	1.96
	± 0.01	± 0.08	± 0.12	± 0.22	± 0.11	± 0.06	± 0.69	± 0.28	± 0.35	± 0.21	± 0.12	± 0.14	± 0.19
Lion	3.39	3.24	4.58	2.95	4.56	4.22	4.93	3.75	5.42	3.13	3.24	1.60	2.70
	± 0.17	± 0.02	± 0.15	± 0.17	± 0.17	± 0.22	± 0.35	± 0.21	± 0.07	± 0.14	± 0.52	± 0.00	± 0.33
Troll	3.95	2.05	4.75	3.17	3.67	3.73	3.82	3.47	4.79	2.71	2.46	1.22	2.11
	± 0.41	± 0.15	± 0.37	± 0.17	± 0.50	± 0.23	± 0.49	± 0.14	± 0.42	± 0.07	± 0.34	± 0.28	± 0.39
Bison_N88	4.88	2.52	5.07	3.38	3.55	4.33	4.93	4.79	6.04	4.03	3.31	1.33	2.03
	± 0.31	± 0.08	± 0.47	± 0.34	± 0.34	± 0.06	± 0.28	± 0.42	± 0.42	± 0.21	± 0.13	± 0.52	± 0.47
Lion_N88	4.29	2.60	4.17	2.95	4.55	4.17	5.07	5.69	6.53	4.10	3.02	1.62	2.89
	± 0.25	± 0.59	± 0.71	± 0.11	± 0.34	± 0.39	± 0.63	± 0.07	± 0.07	± 0.14	± 0.23	± 0.20	± 0.67
Troll_N88	4.27	2.33	4.77	2.83	3.11	4.39	5.42	4.86	5.76	4.10	2.50	1.80	2.51
	± 0.11	± 0.14	± 0.08	± 0.50	± 0.28	± 0.28	± 0.28	± 0.35	± 0.21	± 0.21	± 0.13	± 0.03	± 0.04

Treat- ment	Gross 2022	Gross 2023	Gross 2024	Novi 2022	Novi 2023	Novi 2024	Re 2022	Re 2023	Seegr 2022	Seegr 2023	Vil- bel 2022	Vil- bel 2023	Vil- bel 2024
Lun- abor	2.86 ± 0.15	0.22 ± 0.08	0.62 ± 0.43	0.22 ± 0.00	2.50 ± 0.06	0.78 ± 0.00	3.13 ± 0.14	2.22 ± 1.32	3.54 ± 0.42	3.33 ± 0.07	2.44 ± 0.20	1.39 ± 0.08	0.33 ± 0.15
Pro- bor	2.25 ± 0.15	0.10 ± 0.05	0.35 ± 0.14	0.39 ± 0.05	1.50 ± 0.22	0.83 ± 0.16	3.61 ± 0.00	1.81 ± 1.25	3.19 ± 0.21	3.33 ± 0.42	2.26 ± 0.10	0.88 ± 0.12	0.43 ± 0.15
Jow- isz	2.47 ± 0.13	0.40 ± 0.05	0.67 ± 0.46	0.28 ± 0.06	1.73 ± 0.11	0.89 ± 0.22	3.26 ± 0.14	1.39 ± 0.14	3.68 ± 0.14	3.19 ± 0.07	2.06 ± 0.10	1.02 ± 0.07	0.38 ± 0.12
Bison ×	3.49 ± 0.08	0.79 ± 0.33	3.18 ± 0.33	1.56 ± 0.33	2.89 ± 0.17	2.33 ± 0.11	4.72 ± 0.07	3.47 ± 0.56	4.31 ± 0.28	4.17 ± 0.07	2.73 ± 0.07	1.47 ± 0.27	1.19 ± 0.21
Lun- abor	3.16 ± 0.16	1.01 ± 0.27	2.87 ± 0.30	1.61 ± 0.12	2.45 ± 0.05	2.61 ± 0.11	3.82 ± 0.35	2.64 ± 0.07	4.03 ± 0.00	4.03 ± 0.07	2.72 ± 0.08	1.17 ± 0.08	0.70 ± 0.28
Pro- bor	3.02 ± 0.16	0.63 ± 0.00	3.44 ± 0.06	1.55 ± 0.16	2.62 ± 0.17	2.38 ± 0.11	3.61 ± 0.14	2.50 ± 0.35	4.51 ± 0.14	3.47 ± 0.21	2.69 ± 0.09	0.88 ± 0.25	0.83 ± 0.21
Jow- isz	3.24 ± 0.18	1.21 ± 0.14	3.20 ± 0.09	1.33 ± 0.00	2.95 ± 0.22	2.39 ± 0.11	4.03 ± 0.35	3.96 ± 0.07	4.79 ± 0.14	3.96 ± 0.07	3.11 ± 0.24	1.65 ± 0.06	1.17 ± 0.36
Lun- abor	2.90 ± 0.05	0.90 ± 0.15	1.22 ± 0.00	1.50 ± 0.05	2.55 ± 0.17	2.39 ± 0.17	4.17 ± 0.14	3.19 ± 0.56	4.17 ± 0.00	4.10 ± 0.28	2.69 ± 0.12	1.47 ± 0.20	1.28 ± 0.28
Pro- bor	3.08 ± 0.12	1.09 ± 0.17	3.19 ± 0.00	1.44 ± 0.00	2.94 ± 0.11	2.17 ± 0.11	4.24 ± 0.21	2.99 ± 0.28	4.65 ± 0.14	3.75 ± 0.07	2.73 ± 0.03	0.93 ± 0.03	1.11 ± 0.37
Jow- isz	3.22 ± 0.12	0.68 ± 0.19	3.44 ± 0.45	1.17 ± 0.22	2.89 ± 0.00	2.23 ± 0.00	4.51 ± 0.07	4.17 ± 0.69	4.51 ± 0.21	3.96 ± 0.14	2.56 ± 0.06	1.73 ± 0.02	0.95 ± 0.36
Lun- abor	3.09 ± 0.15	0.43 ± 0.21	1.00 ± 0.68	1.17 ± 0.06	2.06 ± 0.39	1.89 ± 0.12	4.31 ± 0.14	3.54 ± 0.56	4.17 ± 0.35	3.96 ± 0.14	2.50 ± 0.10	1.35 ± 0.18	0.67 ± 0.11
Pro- bor	2.74 ± 0.13	0.55 ± 0.42	3.36 ± 0.41	1.22 ± 0.11	2.56 ± 0.17	2.11 ± 0.16	3.68 ± 0.35	3.26 ± 0.49	4.44 ± 0.07	3.47 ± 0.28	2.54 ± 0.09	1.03 ± 0.18	0.87 ± 0.07
Jow- isz													

Table 13.3

Table S2: Statistical models deployed to analyse the data. The characteristics explain the details of the model. Each response variable (attribute) in the present study was analysed with one or more of the models listed in the table. To analyse the effect of canopy height difference onto the narrow-leaved lupin proportion and the total yield, model M3 was adapted replacing the predictor treatment with canopy height difference.

Table 13.5

Table S3: Year $\times$ location median yields and median absolute deviations (tha^{-1}) aggregated across blocks.

Type	Code	Y	PC1	PC2	PC3	PC4	WAAS	PctRes	Pct- WAAS	wRes	wWAAS	Or- Resp	Or- WAAS
GEN	10	3.0	0.081	0.270	-	0.069	0.081	47.0	93.0	50	50	6	6
					0.250								
GEN	11	2.9	-	-	0.014	0.030	0.081	40.0	93.0	50	50	9	7
			0.081	0.390									
GEN	12	2.8	-	-	0.066	0.041	0.053	36.0	97.0	50	50	12	3
			0.053	0.530									
GEN	13	3.1	0.084	0.060	-	0.170	0.084	52.0	93.0	50	50	4	9
					0.055								
GEN	14	3.0	0.083	-	0.0059	-	0.083	47.0	93.0	50	50	7	8
				0.140		0.290							
GEN	15	3.0	-	-	-	-	0.063	44.0	96.0	50	50	8	4
			0.063	0.230	0.110	0.059							
GEN	16	3.1	-	0.370	0.014	-	0.048	49.0	98.0	50	50	5	2
			0.048			0.120							
GEN	17	2.8	0.080	0.370	-	-	0.080	38.0	94.0	50	50	11	5
					0.230	0.150							
GEN	18	2.8	-	-	-	0.061	0.032	39.0	100.0	50	50	10	1
			0.032	0.120	0.075								
GEN	4	4.3	-	-	-	0.200	0.760	95.0	0.0	50	50	2	15
			0.760	0.0005	0.320								
GEN	5	4.4	-	0.019	0.510	-	0.480	100.0	38.0	50	50	1	10
			0.480			0.0085							
GEN	6	4.0	-	0.260	0.190	-	0.490	85.0	37.0	50	50	3	11
			0.490			0.120							
GEN	7	2.1	0.580	-	0.072	0.250	0.580	12.0	26.0	50	50	13	14
				0.035									
GEN	8	1.9	0.560	0.300	0.260	0.150	0.560	4.0	28.0	50	50	14	13
GEN	9	1.8	0.550	-	-	-	0.550	0.0	29.0	50	50	15	12
				0.210	0.097	0.240							
ENV	Gross	2.7	-	-	-	0.120	0.600	43.0	41.0	50	50	3	3
			0.600	0.110	0.600								

Table 13.7

Table S4: The additive main effects and multiplicative interaction (AMMI) model output

Type	Code	Y	PC1	PC2	PC3	PC4	WAAS	PctRes	Pct- WAAS	wRes	wWAAS	Or- Resp	Or- WAAS
ENV	Novi	2.5	-	-	0.500	0.058	0.740	33.0	26.0	50	50	4	4
			0.740	0.370									
ENV	Re	3.9	-	0.910	0.120	-	0.058	88.0	100.0	50	50	2	1
			0.058			0.015							
ENV	Seegr	4.2	0.420	-	-	-	0.420	100.0	61.0	50	50	1	2
				0.230	0.065	0.490							
ENV	Vil-	1.7	0.980	-	0.045	0.320	0.980	0.0	0.0	50	50	5	5
	bel			0.200									

Appendix A: Hypothesis analysis.

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